Lesson Notes

MODULE 3 | LESSON 3

SOME ABUSES OF THE BLACK-LITTERMAN MODEL

Reading Time	60 minutes
Prior Knowledge	Black-Litterman model. Some understanding of Bayesian statistics is useful as well.
Keywords	Black-Litterman without equilibrium model, Black-Litterman with data based prior, quadratic utility and loss of utility.

In the previous lesson we studied the critical line algorithm which is an excellent mathematical tool to compute optimal portfolios, both in the traditional setting and in the Black-Litterman approach. In this lesson we will explore some situations where, if the set-up of the Black-Litterman model is not rigorous, one could end up with suboptimal results.

1. Generalities

By now, students are acquainted with the Black-Litterman model and are aware of at least a few techniques that can be used to implement the model in practice. In reality, those techniques apply to Black-Litterman and the classic mean-variance optimization problem alike.

Practitioners are often driven by a certain urgency to avoid complications or look for short-cuts and for such reasons have developed approaches that are not necessarily rigorous.

While these approaches to the Black-Litterman model are good to cut corners, so to speak and tend to work overall, they lead to some loss of precision. This particular topic is the subject of this lesson.

This lesson in based on the mandatory reading, "Uses and misuses of the Black-Litterman model in portfolio construction".The paper describes 3 instances were the implementation of the Black-Litterman model lead to suboptimal solutions.

In the notes for this lesson we will cover some mathematical aspects which are basically assumed in the paper hoping that every student can get as much as possible from the reading itself.

2. Not Using an Equilibrium Model

In Lesson 1 of this Module we covered the Black-Litterman model as intended by its authors: i.e., investors' subjective views are combined with an equilibrium model. Essentially, the portfolio weights are anchored around the implied market capitalization weights and are allowed to move away from those in amounts that depend on the investors uncertainty about their own views. The less convinced an investors or portfolio manager is about his or her views (i.e. higher variance around their mid-point view) the more they will tend to stick close to what we can call the market's view or the equilibrium view.

There is an approach to Black-Litterman in which we dispense with the equilibrium model. To this end, the authors mention a model proposed by Jones, Lim, and Zangari in 2007 and that they call the **JLZ model** and which is a factor-based model. This model requires
the following steps:

Step 1. Identify the factor model by identifying K factors. Then for a portfolio consisting of N assets, we let B be an $N \times K$ matrix that represents factor exposure as we saw in Lesson 4 of Module 2. Hence the portfolio exposure to these factors overall is given by $B^T w$ with w an $N \times 1$ vector of weights.

Step 2. A portfolio manager specifies factor views using a$K \times 1$ vector λ such that $B^T w = \lambda$.

Step 3. The optimal weight that is optimal given the factor view vector λ can now be determined by solving the following problem:

$$\min_{w} w^T \Sigma w$$
$$\text{subject to : } B^T w = \lambda.$$

Now,Σ is the unknown variance covariance matrix of the returns of the assets which is unknown. However this is not a problem as we can estimate it as usual, i.e. we compute $\widehat{\Sigma}$. Then the optimization problem can be run as

$$\min_{w} w^T \widehat{\Sigma} w$$
$$\text{subject to : } B^T w = \lambda.$$

We should also notice that there is no reference to the expected returns μ. If we solve the optimization program, i.e. we form the Lagrangian function

$$L = \frac{1}{2} w \widehat{\Sigma} w + \delta^T (\lambda - B^T w)$$

then the first order conditions, obtained by setting the first derivatives with respect to w and δ equal to zero yield

$$\widehat{\Sigma} w - B\delta = 0 \quad \text{and} \quad B^T w = \lambda.$$

(Pay attention that in our formulation,δ is not a scalar but a $K \times 1$ vector since the constraints are in matrix form and contain actually K constraints).

Now,

$$w = \widehat{\Sigma}^{-1} B \delta$$

$$\delta = (B^T \widehat{\Sigma}^{-1} B)^{-1} \lambda$$

and combining these two sets of equations we find

$$w_{opt} = \widehat{\Sigma}^{-1} B (B^T \widehat{\Sigma}^{-1} B)^{-1} \lambda$$

which is formula (2) in the required reading.

Step 4. At this point we need to compute what the paper calls the Black-Litterman α by solving the problem

$$w_{opt} = \mathbf{argmax}\ w^T \alpha - \frac{1}{2} w^T \widehat{\Sigma} w.$$

If we compute the partial derivatives of the objective function on the right-had side of the equal sign with respect to w and set it equal to zero, then

$$\alpha - \widehat{\Sigma} w = 0$$

yields

$$w = \widehat{\Sigma}^{-1} \alpha$$

and now we use w_{opt} in place of w and we find

$$\alpha = \widehat{\Sigma} w_{opt}$$

which is (4) in the assigned reading. We call this α that was just found α_{BL}.

Step 5. This step is similar to what we have already seen several times already, i.e. we solve the following quadratic problem:

$$\begin{aligned} &\max_w w_{BL}^{\alpha} - \frac{1}{2} w^T \widehat{\Sigma} w \\ &\textbf{subject to}: w^T \iota = 1 \\ &\qquad\qquad l_i \leq w_i \leq u_i, \ \ i = 1, 2, \ldots, N. \end{aligned}$$

which we can solve using gradient based algorithms of the critical line algorithm learned in the lesson before this one and where l_i and u_i represent lower and upper bounds for the weights of asset i for $i = 1, 2, \ldots, N$.

The paper explains that **we are not using an equilibrium model because the data are used to compute $\widehat{\sum}$ but do not use the information about the expected returns μ.** Hence, in the parlance of the Black-Litterman model, subjective views are combined with the model, but only partially as we are ignoring μ.

Then, the authors proceed to analyze what is the impact of such approach when compared to the true Black-Litterman methodology. To do so, they introduce a quadratic utility function

$$U(\mu, \sigma) = \mu - \frac{\delta}{2} \sigma^2$$

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similar to some of those we saw in Module 1 and in Module 2 with μ the mean return of the portfolio and σ the volatility of the portfolio. Then the value of the utility is computed both using the Black-Litterman model and the JLZ model to assess how much utility was lost in the process.

To achieve this, investors' views must be formalized in a probability model and the authors consider:

$$r_{t+1} = B_t f + e_{t+1}, \quad e_{t+1} \sim N(0, (1/h)V_e).$$

In addition, we have also

$$f|h \sim N(m_f, (1/h)V_f)$$

$$h \sim \text{Gamma}(p, q).$$

The choice of using a Gamma distribution is due to the fact that it is a conjugate prior and thus makes the computations much simpler since the posterior distribution would be of the same type as the prior distribution, i.e., multivariate normal.

The paper provides equation (9) as the formula for λ and it also contains a generous note/comment (footnote 13). Formula (9) is a consequence of the use of the hierarchical model involving normal random variables that was selected. In fact, if we start from

$$r_{t+1} = B_t f + e_{t+1}, \quad e_{t+1} \sim N(0, (1/h)V_e), \quad f|h \sim N(m_f, (1/h)V_f)$$

one can prove that the marginal distribution of r_{t+1} is normal.

The easiest way do it is arguably via characteristic functions or moment generating functions paired with the law to total expectation. In fact, in general, if we are given that

$$X|Y = y \sim N(\alpha y, \sigma^2) \quad \text{and} \quad Y \sim N(\mu, \delta^2),$$

then the moment generating function for X can be computed using the definition and the law of total probability, i.e.:

$$\phi_X(t) = E[e^{tX}] = E_Y[E_{X|Y}[e^{tX}|Y]] = E_Y[e^{t\alpha Y + \frac{t^2\sigma^2}{2}}]$$

and now we can compute this second expectation exploiting the fact $Y \sim N(\mu, \delta^2)$. This is not a difficult calculation, albeit a bit tedious since one has to complete the square in the exponent for the exponential, etc. In any case, the calculation yields

$$\phi_X(t) = e^{\alpha\mu t + (\alpha^2\delta^2 + \sigma^2)\frac{t^2}{2}},$$

which is recognizable as the moment generating function of a normal random variable with a certain mean and a certain variance.

The unconditional mean and variance for our variables, which is what we need, are not terribly complicated to compute. In fact, using again the law of total expectation, we have

$$E[r_{T+1}] = E[E[r_{T+1}|f]] = E[B_T f] = B_T E[f] = B_T m_f.$$

For the unconditional variance it is a bit more work but one uses the well-known formula:

$$\mathbf{Var}(X) = E[\mathbf{Var}(X|Y)] + \mathbf{Var}(E]X|Y])$$

which in our case yields

$$\mathbf{Var}(r_{T+1}) = E[\mathbf{Var}(r_{T+1} \mid f)] + \mathbf{Var}(E]r_{T+1}|f])$$

and

$$E[\mathbf{Var}(r_{T+1} \mid f)] = E[B^T(1/h)V_f B_T^T + (1/h)V_e] = B^T(1/h)V_f B_T^T = E[(1/h)V_e] \equiv \Omega.$$

It is not difficult to show that

$$\mathbf{Var}(E]r_{T+1}|f]) = 0.$$

Hence we have shown that r_{t+1} has the following unconditional distribution.

$$r_{T+1} \sim N(B_T f, \Omega).$$

If we now run the optimization program

$$\max_w w^T(B_T f) - \frac{1}{2} w^T \Omega w$$

we find

$$w_{opt}^{\times} = \Omega^{-1} B_T m_f$$

and finally solving for $\lambda = B_T^T w_{opt}^{\times}$ we get

$$\lambda = B_T^T w_{opt}^{\times} = B_T^T \Omega^{-1} B_T m_f$$

which is indeed (9).

Now, one has everything to start predicting. The process is well described in the reading at page 156 (a 7 step process). The authors need to derive the predictive expectation and the predictive variance-covariance matrix (formulae (14), (15), and (16)). These are quite complex and presented without proof. In fairness, the proof is not particularly difficult in terms of mathematics; however it is very long, like 4 pages of algebraic calculations (it is not an unusual occurrence when dealing with Bayesian methods). The proof is contained in the original paper submitted for publication and can be obtained from the authors if students are interested (a reference will be provided at the end of the lesson). Equipped with the predictive mean and variance-covariance, it is shown that the difference in utility between two investors having the same views, but where one uses the information efficiently (RTL) and the other uses the JLZ model, is

$$U_{RTL} - U_{JLZ} = \frac{1}{2}(m_f - \tilde{E}[f])^T B_{T+1} \tilde{V}(r_{T+1})^{-1} B_{T+1}(m_f - \tilde{E}[f])$$

and it is clearly a positive number since the loss is represented by a quadratic form with positive coefficient. The key element is that the loss depends on the magnitude of $m_f - \tilde{E}[f]$, the difference between the prior mean and the the posterior mean of the factor premium.

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3. Data Based Prior

In this case, what the authors refer to the model as the **FFK model**, which utilizes a factor model to represent the portfolio manager's views. That is, we start with a factor model

$$r_{t+1} = B_t f + e_{t+1}, \qquad \epsilon_{t+1} \sim N(0, \Sigma_e)$$

where in this case the unknown are the value of the factor and the variance covariance matrix Σ_e. $Sigma_e$ represents the uncertainty about the view for each factor and, as described in Lesson 1 of this module discussing the Black-Litterman model the views are independent and therefore Σ_e is a diagonal square matrix.

These quantities can be estimated using OLS or, more likely, GLS/WLS techniques as explained in Lesson 4 of Module 2. In this way we will have two estimates $\widehat{f}$ and $\widehat{\Sigma}_e$. Now, the portfolio manager can express his or her prior for the mean return $E[r_{T+1}]$ in the form

$$g \sim N(B_T \widehat{f}, \widehat{\Sigma}_e)$$

Note that this model is *just a belief that a factor model is the underlying data generating process*. However, in this approach one also assumes that π is the estimated equilibrium return obtained via the CAPM and that

$$\pi = g + \epsilon, \;\; \epsilon \sim N(0, \tau\Sigma)$$

where in this case Σ is the variance-covariance matrix of the future returns r_{T+1}

This can be estimated and we can write $\pi = \widehat{\Sigma} w_M$. Then from the definition of π above, we can estimate g and $\mathbf{Var}(g)$; specifically:

$$\hat{g} = \pi = \widehat{\Sigma} w_M, \;\; \widehat{\mathrm{Var}}(\hat{g}) = \tau\widehat{\Sigma}$$

From this, deriving equation (31) in the paper is not exactly complicated, but calculations are a bit tricky.

Using the Bayesian setting for posterior distributions, we have that

$$f_{g|\pi} = \frac{k}{f_\pi} \cdot \exp\left\{ -\frac{1}{2}(\pi - g)^T (\tau\widehat{\Sigma})^{-1} (\pi - g) - \frac{1}{2}(g - B_T\widehat{f})^T (\widehat{\Sigma}_e)^{-1} (g - B_T\widehat{f}) \right\}$$

where k is just a constant of proportionality and f_π is the marginal density for π.

Up to this point the step was relatively straightforward. However, to simplify this expression, one has to introduce some algebraic tricks. Here we follow the process delineated in Satchell and Scowcroft (reference provided at the end of the Lesson). Then, if we let H be the exponent of the exponential in the formula above, i.e.:

$$H \equiv -\frac{1}{2}(\pi - g)^T (\tau\widehat{\Sigma})^{-1} (\pi - g) - \frac{1}{2}(g - B_T\widehat{f})^T (\widehat{\Sigma}_e)^{-1} (g - B_T\widehat{f}),$$

with some simple algebra, we find that

$$H = g^T(\tau\widehat{\Sigma})^{-1}g - 2\pi(\tau\widehat{\Sigma})^{-1}g + \pi^T(\tau\widehat{\Sigma})^{-1}\pi = g^T\widehat{\Sigma}_e^{-1}g + 2(B_T\widehat{f})^T\widehat{\Sigma}_e^{-1}(B_T\widehat{f}).$$

The next step is the somewhat tricky one (but it amounts to just algebra nevertheless) and it is the step described in Satchell and Scowcroft. This step consists in rewriting H as

$$H = g^T M^T M^{-1} M g - 2NM^{-1}Mg + A$$

where

$$M = (\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1},$$

$$N = (\tau\widehat{\Sigma})^{-1}\pi + \widehat{\Sigma}_e^{-1}B_T\widehat{f},$$

$$A = (BT\widehat{F})^T\widehat{\Sigma}_e^{-1}(B_T\widehat{F}) + \pi^T(\tau\widehat{\Sigma})^{-1}\pi,$$

and we should notice that M is symmetrical as it involves variance covariance matrices. Finally, the exponent can be further simplified to look like

$$H = (g - M^{-1}N)^T M (g - M^{-1}C) + A - N^T M^{-1} N.$$

Now, the terms $A - N^T M^{-1} N$ does not contain g and can be absorbed into the constant of integration. Thus, we have found that

$$f_{g|\pi} = k \exp\{(g - M^{-1}N)^T M (g - M^{-1}C)\}.$$

This is very useful for we can recognize that the posterior is a normal distribution whose mean, $\widehat{E}[g],$ is given by

$$M^{-1}N = [(\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1}]^{-1}[(\tau\widehat{\Sigma})^{-1}\pi + \widehat{\Sigma}_e^{-1}B_T\widehat{f}]$$

and the variance is found as

$$\widehat{\mathrm{Var}}[g] = [(\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1}]^{-1},$$

which is (31) as we would have liked to show.

At this point, the relevance of equation (32) is obvious. Now we have a new contradiction: a model where it is assumed that the data is generated by a factor model and a likelihood function that uses a belief according to which the CAPM is the data generating process.

Using the same setup for the previous case, the authors show that the loss in utility between using the true efficient model and the FFK is given by formula (440, p. 162)

$$U_{RTL} - U_{FFK} = -\frac{1}{2}(\widehat{E}[g] - \tilde{E}[r_{T+1}])^T(\widetilde{\mathrm{Var}}[r_{t+1}])^{-1}(\widehat{E}[g] - \tilde{E}[r_{T+1}]).$$

Futhermore, the authors point out that the loss derived from using the model FFK is somewhat smaller than that of using the JLZ model.

4. Reverse Optimization Technique

This part is more straightforward and does not require any particular comment.

5. Conclusion

While the approaches described are simpler and of more immediate application, they are either ignoring part of the information available or using "pieces" from contradictory models (CAPM e factor models). This leads to about a 1% reduction in the estimated mean and, while 1% may not seem like much, we are aware from previous lessons that the estimates for the mean have a big impact on the determination of optimal portfolios.

The third part of the paper explains that in using reverse optimization, practitioners rely on the estimated variance covariance matrix which is estimated with error and such error can often be far from negligible. This should cause practitioners to be very careful and aware of potential misuses of the Black-Litterman model.

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